

Computer Vision SS26 – Exercise 8

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This exercise will be discussed the week of 13th of July.

This exercise sheet covers Multiview Stereo, Plane Sweep Stereo, Differential Rendering, and Neural Radiance Fields.

1 Exercises

Exercise 1: Plane Sweep Stereo

Plane Sweep Stereo estimates depth by evaluating a set of hypothetical depth planes.

1. Explain the idea of Plane Sweep Stereo. Why are neighboring images reprojected onto hypothetical planes?
2. Assume five candidate depth planes

$$z = \{1, 2, 3, 4, 5\} \text{ m.}$$

For one pixel, the matching costs are

$$C(z) = \{0.81, 0.42, 0.18, 0.37, 0.64\}.$$

Determine the estimated depth.

3. Explain how the cost profile of a single pixel becomes a depth map for the complete image.
4. Compare Plane Sweep Stereo with classical block matching for stereo.

State one advantage and one disadvantage of each method.

Exercise 2: Multi-View Stereo

The multiview stereo pipeline consists of

Images $\rightarrow$ Camera poses $\rightarrow$ Dense correspondences $\rightarrow$ Depth maps $\rightarrow$ Depth map fusion $\rightarrow$ 3D model.

1. Explain the purpose of each processing step.
2. Why are several neighboring views used instead of only one stereo pair?
3. Explain why depth-map fusion is necessary before obtaining the final surface.
4. Name two typical sources of reconstruction errors in multiview stereo.

Exercise 3: Differentiable Rendering

Computer vision can be formulated as an inverse rendering problem.

1. Explain the difference between rendering and inverse rendering.
2. Why must the rendering process be differentiable for optimization using gradient descent?
3. Assume an observed pixel has intensity

$$I_{\text{obs}} = 0.72,$$

while the rendered image predicts

$$I_{\text{pred}} = 0.63.$$

Using the squared reconstruction loss

$$L = (I_{\text{pred}} - I_{\text{obs}})^2,$$

compute the loss.

4. Explain how minimizing this loss can improve the estimated scene representation.

Exercise 4: Neural Radiance Fields (NeRF)

A Neural Radiance Field represents a scene by a neural network

$$F(\mathbf{x}, \mathbf{d}) \longrightarrow (\sigma, \mathbf{c}),$$

where

- $\mathbf{x}$ denotes a 3D position,
- $\mathbf{d}$ denotes the viewing direction,
- σ is the volume density,
- $\mathbf{c}$ is the emitted RGB color.

1. Explain the meaning of the inputs $\mathbf{x}$ and $\mathbf{d}$.
2. Explain the meaning of the outputs σ and $\mathbf{c}$.
3. Why does NeRF predict color as a function of viewing direction?
4. Compare NeRF with classical multiview stereo. Discuss one advantage and one disadvantage of each approach.